Structures & Records in C++

Lab 03 Solutions: Inventory System & Carpet Cost Calculator

Object Oriented Programming

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Task 01 (Part 1): Inventory with int code

```
#include <iostream>
#include <iomanip>
#include <cstdlib>
#include <ctime>
#include <string>
using namespace std;

const int SIZE = 10;

struct InventoryItem
{
    int code;
    string name;
    double cost;
    double price;
};

InventoryItem insertItem()
{
    InventoryItem item;
    item.code = 3123 + rand() % 9000;                // auto code
    cout << "Item name: ";
    getline(cin >> ws, item.name);
    cout << "Item cost: ";
    cin >> item.cost;

    item.price = item.cost * 1.40;                   // 40% profit
    return item;
}

void displayStock(const InventoryItem* item)
{
    cout << left << setw(8) << item->code
         << setw(18) << item->name
         << setw(12) << item->cost
         << setw(12) << item->price << endl;
}

int main()
{
    srand(time(0));
    InventoryItem stock[SIZE];
```

```
    for (int i = 0; i < SIZE; i++)
        stock[i] = insertItem();

    cout << left << setw(8) << "Code"
         << setw(18) << "Name"
         << setw(12) << "Cost"
         << setw(12) << "Price" << endl;
    cout << string(50, '-') << endl;

    for (int i = 0; i < SIZE; i++)
        displayStock(&stock[i]);
    return 0;
}
```

Task 01 (Part 2): Code as string (e.g. COM_31203)

Same as Part 1, but code is a string: first 3 letters (upper case) + _ + running number.

```
#include <iostream>
#include <iomanip>
#include <cctype>
#include <string>
using namespace std;

const int SIZE = 10;
int counter = 31203;          // running integer

struct InventoryItem
{
    string code;              // now a string
    string name;
    double cost;
    double price;
};

// builds code like COM_31203
string makeCode(const string& name)
{
    string pre = name.substr(0, 3);
    for (int i = 0; i < (int)pre.size(); i++)
        pre[i] = toupper(pre[i]);
    return pre + "_" + to_string(counter++);
}

InventoryItem insertItem()
{
    InventoryItem item;
    cout << "Item name: ";
    getline(cin >> ws, item.name);
    cout << "Item cost: ";
    cin >> item.cost;

    item.code = makeCode(item.name);
    item.price = item.cost * 1.40;
    return item;
}

void displayStock(const InventoryItem* item)
{
    cout << left << setw(12) << item->code
         << setw(18) << item->name
         << setw(12) << item->cost
```

```
          << setw(12) << item->price << endl;
}

int main()
{
    InventoryItem stock[SIZE];
    for (int i = 0; i < SIZE; i++)
        stock[i] = insertItem();

    cout << left << setw(12) << "Code"
         << setw(18) << "Name"
         << setw(12) << "Cost"
         << setw(12) << "Price" << endl;
    cout << string(54, '-') << endl;

    for (int i = 0; i < SIZE; i++)
        displayStock(&stock[i]);
    return 0;
}
```

Task 02: Carpet Cost

Price per sq ft = Rs. 18. Inches are converted to feet (inches / 12).

```
#include <iostream>
using namespace std;

const int PRICE_PER_SQFT = 18;              // Rs. per sq ft

struct FeetInches
{
    int feet;
    int inches;
};

struct RoomDimension
{
    FeetInches length;
    FeetInches Width;
};

// converts feet + inches to feet
double toFeet(const FeetInches& f)
{
    return f.feet + f.inches / 12.0;
}
```

```
// cost of carpet for one room
int computeTheCarpetingCost(const RoomDimension& room)
{
    double area = toFeet(room.length) * toFeet(room.Width);
    return (int)(area * PRICE_PER_SQFT);
}

int main()
{
    int n;
    cout << "Number of rooms: ";
    cin >> n;

    RoomDimension* rooms = new RoomDimension[n];
    for (int i = 0; i < n; i++)
    {
        cout << "Room " << i + 1 << endl;
        cout << " Length (feet inches): ";
        cin >> rooms[i].length.feet >> rooms[i].length.inches;
        cout << " Width (feet inches): ";
        cin >> rooms[i].Width.feet >> rooms[i].Width.inches;
    }

    int choice;
    do
    {
        cout << "\n1. Total cost of house\n"
             << "2. Cost of each room\n"
             << "0. Exit\n"
             << "Choice: ";
        cin >> choice;

        if (choice == 1)
        {
            int total = 0;
            for (int i = 0; i < n; i++)
                total += computeTheCarpetingCost(rooms[i]);
            cout << "Total cost: Rs. " << total << endl;
        }
        else if (choice == 2)
        {
            for (int i = 0; i < n; i++)
                cout << "Room " << i + 1 << ": Rs. "
                     << computeTheCarpetingCost(rooms[i]) << endl;
        }
    } while (choice != 0);

    delete[] rooms;
    rooms = nullptr;
    return 0;
}
```